Exercise Sheet

Lecture 9

Adjoints; Self-Adjoint, Unitary, and Normal Operators

Inner products are in the physics convention (conjugate-linear in the first slot); the adjoint satisfies $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. Tags mark the flavour; a ⋆ marks a harder one.

Core tutorial route

Work **9.1, 9.4, 9.5, 9.9, 9.12, 9.15, 9.19, and 9.22** in that order (about 80–100 minutes before discussion). This eight-problem route covers adjoint computation and algebra, a real self-adjoint diagonalization, the real-spectrum proof, a guided unit-circle proof, normal-only classification, and the normal length/shared-adjoint/orthogonality chain.

Additional practice: 9.2–9.3, 9.6–9.8, 9.10–9.11, 9.13–9.14, 9.16–9.18, 9.20–9.21. **Bridge:** none. **Extension:** 9.23–9.24 (commutators and complex skew-adjoint spectra). Extension tasks do not count toward core progress.

Adjoints

Exercise 9.1 (computational) Find the adjoint of $A = \begin{pmatrix} 2 & 1+i \\ 3 & -i \end{pmatrix}$, and state whether A is self-adjoint, unitary, or normal.

Answer. $A^\dagger = \begin{pmatrix} 2 & 3 \\ 1-i & i \end{pmatrix}$; not self-adjoint and not normal (so not unitary).

Exercise 9.2 (computational) On $\mathbb{R}^2$ with the standard inner product, find the adjoint of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, and say whether A is self-adjoint.

Answer. $A^\dagger = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$; not self-adjoint.

Exercise 9.3 (computational) Find the adjoint of $A = \begin{pmatrix} i & 2 \\ 1-i & 3i \end{pmatrix}$ on $\mathbb{C}^2$.

Answer. $A^\dagger = \begin{pmatrix} -i & 1+i \\ 2 & -3i \end{pmatrix}$.

Exercise 9.4 (computational) Let $T(z_1, z_2) = (2z_1 + iz_2,\ z_1 - z_2)$ on $\mathbb{C}^2$. Find $T^\dagger$ as a formula.

Answer. $T^\dagger(z_1, z_2) = (2z_1 + z_2,\ -iz_1 - z_2)$.

Exercise 9.5 (proof) Prove that $(ST)^\dagger = T^\dagger S^\dagger$ for all $S, T \in \mathcal{L}(V)$.

Exercise 9.6 (proof) Prove that null $T^\dagger = (\text{range}\, T)^\perp$.

Exercise 9.7 (proof) Prove that for every $T \in \mathcal{L}(V)$ both $T + T^\dagger$ and $T^\dagger T$ are self-adjoint.

Self-adjoint operators

Exercise 9.8 (computational) Verify that $A = \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix}$ is Hermitian, and explain why its eigenvalues must be real.

Answer. $A^\dagger = A$, so A is Hermitian; hence its eigenvalues are real.

Exercise 9.9 (computational) Verify that $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ is self-adjoint on $\mathbb{R}^2$, and find its eigenvalues and an orthonormal eigenbasis.

Answer. Symmetric, so self-adjoint; eigenvalues $5, 3$ with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$.

Exercise 9.10 (computational) Check that $A = \begin{pmatrix} 3 & i \\ -i & 3 \end{pmatrix}$ is Hermitian and find its (real) eigenvalues and corresponding eigenvectors.

Answer. $A^\dagger = A$; eigenvalues $4, 2$ with eigenvectors $(i, 1)$, $(-i, 1)$ (orthogonal).

Exercise 9.11 (computational) Find all $a \in \mathbb{C}$ for which $A = \begin{pmatrix} 5 & a \\ 3-2i & -1 \end{pmatrix}$ is self-adjoint.

Answer. $a = 3 + 2i$.

Exercise 9.12 (proof) Prove that the eigenvalues of a self-adjoint operator are real, directly from $Tv = \lambda v$.

Exercise 9.13 (proof) Suppose T is self-adjoint and $Tu = \alpha u$, $Tv = \beta v$ with $\alpha \neq \beta$. Prove $\langle u, v \rangle = 0$.

Exercise 9.14 (proof) Suppose T is self-adjoint and $T^2 = 0$. Prove that $T = 0$.

Unitary and normal operators

Exercise 9.15 (proof) Show that $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ is unitary. Then, starting from $Uv = \lambda v$ with $v \neq 0$, prove directly that every eigenvalue of U lies on the unit circle.

Exercise 9.16 (proof) Prove that a unitary operator preserves the inner product: $\langle Uu, Uv \rangle = \langle u, v \rangle$.

Exercise 9.17 (computational) Let $S(w, z) = (-z, w)$ on $\mathbb{C}^2$. Find $S^\dagger$, show S is normal, and find its eigenvalues.

Answer. $S^\dagger = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -S$; S is normal (in fact unitary), eigenvalues $\pm i$.

Exercise 9.18 (computational) Show that $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ is unitary, and state $|\lambda|$ for its eigenvalues.

Answer. $Q^\dagger Q = I$, so Q is unitary; every eigenvalue has $|\lambda| = 1$.

Exercise 9.19 (computational) Determine whether $A = \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$ is normal, self-adjoint, or unitary, and find its eigenvalues.

Answer. Normal (indeed $A^\dagger A = AA^\dagger = 13I$); neither self-adjoint nor unitary; eigenvalues $2 \pm 3i$.

Exercise 9.20 (computational) Classify $D = \operatorname{diag}(i, -i, 1)$ on $\mathbb{C}^3$ as self-adjoint, unitary, and/or normal.

Answer. Normal and unitary, but not self-adjoint.

Exercise 9.21 (computational) Show that $A = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ is normal, and find its eigenvalues. Is it self-adjoint or unitary?

Answer. $A^\dagger A = AA^\dagger = 2I$, so A is normal; eigenvalues $1 \pm i$; neither self-adjoint nor unitary.

Exercise 9.22 ($\star$) Prove that T is normal if and only if $\|Tv\| = \|T^\dagger v\|$ for all v. Then prove the shared-adjoint statement $Tv = \lambda v \iff T^\dagger v = \overline{\lambda} v$, and use it to prove that eigenvectors for distinct eigenvalues of a normal operator are orthogonal.

Hint. First rewrite the two squared norms as quadratic forms. Next apply the length identity to $T - \lambda I$. For orthogonality, move T across the inner product.

Exercise 9.23 ($\star$) On a complex inner-product space, suppose A, B are self-adjoint. Prove AB is self-adjoint if and only if $AB = BA$, and that $i(AB - BA)$ is always self-adjoint.

Hint. Use $(AB)^\dagger = B^\dagger A^\dagger = BA$. For the commutator, take the adjoint of $i(AB - BA)$ and track the sign.

Exercise 9.24 (proof) On a complex inner-product space, an operator is *skew-adjoint* if $B^\dagger = -B$. Prove that every eigenvalue of a skew-adjoint operator is purely imaginary.